Suggested problems: 1, 3, 6, 8

Problem 1: Define the function $\sigma : \mathbb{R} \to \mathbb{R}$ by setting

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

This is called the *sigmoid function*.

(a) Draw a graph of σ.

(b) Compute the range of σ.

Problem 2: Suppose that X_1 and X_2 are independent indicator random variables, both distributed as $\mathcal{B}er(1/2)$, where X_1 codes whether a patient smokes and X_2 codes whether they have family history of a certain disease. Suppose Y is a third indicator random variable that codes whether the patient has the disease. Suppose that we model Y with a logistic regression model with predictors X_1 and X_2, and parameters $\beta_0 = -2$, $\beta_1 = 1$, and $\beta_2 = 1.5$.

(a) Compute the marginal p.m.f. $f(y)$.

(b) Given that a patient has the disease, what is the probability that they both smoke and have a family history?

(c) Given that a patient has the disease, what is the probability that they smoke?